TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-assisted, sensor-based bioreactor system for quality-controlled production of 3D cell models			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-process Technologies Bio-process Equipment Technologies Bio-Analytical Instruments	Robot/automation machines	Automation-related measuring/sensor technology
Concept of Proposed Technology	<u>Core concept</u> We propose an AI-supported, sensor-based bioreactor system for the quality-controlled, scalable production of complex 3D cell models, e.g. those derived from iPSCs. The proposed system is based on non-invasive sensors for process parameters and trained ML models for classifying 3D cell aggregates, both of which are already available at Fraunhofer IBMT. In addition, work on sorting based on micro fluidics of complex 3D cell models already underway at Fraunhofer IBMT will be further advanced and integrated into the overall system. 3D cell models (e.g., spheroids or organoids) exiting the bioreactor are analyzed and sorted in a downstream module to achieve batches of aggregate populations with narrow size distribution. The platform enables automated			

sorting of large-scale 3D cell-model batches during and/or after steps of the bioprocess (e.g., cultivation, differentiation and maturation).

Main functions

The AI-assisted, sensor-based bioreactor system for quality-controlled production of complex iPSC-derived 3D cell models will offer the following capabilities:

- Optimized cultivation and expansion of sensitive cell models using a sensor system comprising: a) Inline sensors for continuous monitoring of process parameters inside the bioreactor (e.g., temperature, CO_2, pH and other culture-relevant metrics). b) Online sensors for continuous, non-invasive optical quality control of produced 3D cell aggregates. c) At-line sensors/actuators for AI-assisted classification of 3D cell aggregates by target parameters (e.g., size, morphology) and high-throughput sorting according to these parameters.
- Parallelization and partial automation of 3D cell model production in scalable formats, including automated medium exchange driven by sensor data in real-time.
- AI-supported, non-invasive assessment of cell cultures during production, with potential online adaptation and optimization of process conditions.
- Sterile, high-throughput sorting directly after and based on AI-based characterisation to produce homogeneous 3D cell-model batches with controlled size distribution and morphology.
- Data integration and interoperability with Quality Systems (QS) via appropriate interfaces.

Potential uses of the proposed technology

The proposed technology targets growth markets in 3D cell culture, bioreactors, iPSC technology, spheroids, and organoids. A subsequent exploitation path involves a Korean SME, supported by Fraunhofer IBMT, developing a **custom device for a SME-defined application**, marketing it within Korea, and, if possible, adapting the platform to multiple product areas. Potential applications include:

- **Pharma production**: quality-controlled manufacturing of diverse iPSC-derived products and advanced therapy medicinal products (ATMPs).
- **Drug discovery and chemical evaluation**: AI-driven, complex process workflows for monitoring, quality assurance, and elucidating relationships in biomedical studies (e.g., drug screening, drug safety, toxicity testing, toxicity of chemical compounds).
- **Bioprocess and Lab-Automation**: development of modular automation solutions for biotechnology with standardized interfaces for digital networking.
- **Cell culture automation**: adaption of the system for various cell systems (e.g., pluripotent stem cells, mesenchymal stem cells, primary cells, immortalized cell lines)

For valorization, the project could deliver either:

- The individual components (a) inline bioreactor sensors, (b) online quality-control sensors for 3D aggregates, (c) at-line AI-driven classification and sorting with high-throughput capability; or
- A combined, customized complete system assembled from these components, adapted to a specific application and subsequently marketed as a turnkey solution.

Rationale for Joint R&D Project	In addition, Korean SMEs will benefit from the expertise of Fraunhofer IBMT to tailor the technology to their research or production needs, enabling in-country uptake. <u>Need for this joint R&D project</u> Human stem cell-based 3D cell models more accurately recapitulate organ architecture and physiology than traditional 2D models. Achieving high-throughput, high-quality production of complex iPSC-derived cell models – ensuring homogeneous and reproducible batches in terms of size and morphology – is essential for expanding the application portfolio of these models in drug development and as potential alternatives to animal testing. Current tools and processes lack a fully automated, end-to-end production chain that integrates real-time monitoring, quality assurance, and high-throughput sorting. <u>Why collaboration is required</u> • To scale the production of iPSC-derived 3D cell models, an integrated platform combining inline sensing, online quality assessment, AI-driven at-line classification, and sterile sorting is required. This integration spans biosensing, fluidics, AI, and bioprocess automation disciplines. • Synergies between Fraunhofer IBMT and the Korean SME will drive development: the SME gains access to Fraunhofer IBMT's networks and expertise, while Fraunhofer IBMT benefits from the Korean market and new application areas for its technology. • Fraunhofer IBMT brings foundational research, algorithmic development, and system integration expertise for AI-based monitoring, non-invasive assessment, and data interoperability with QS systems. A Korean SME is essential to tailor the device to specific application relevant to Korea, to assist in application-specific connection to a QS system, provide local manufacturing capabilities, regulatory knowledge, and market access, and lead commercialization in Korea. • Fraunhofer IBMT cannot market or sell products; a local SME partner is needed to translate the technology into a market-ready device and to establish in-market deployment. The proposed joint project enables knowledge transfer, reduces time-to-market, and creates a commercially viable platform with cross-border collaboration.
Key Words	1. Quality-assured production of 3D cell models 2. Comprehensive process control für optimized culture of 3D cell models 3. AI-assisted sensor-based bioreactor 4. Inline and online non-invasive quality control 5. Fluidics-based, AI-assisted sorting of 3D cell aggregates on the production line (at-line)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	• Develop an integrated, AI-assisted sensor-based bioreactor platform for high-quality, scalable production of complex iPSC-derived 3D cell models, including inline process sensing, online non-invasive quality control, and at-line AI-driven sorting. • Create robust AI/models for real-time, non-invasive assessment of 3D cell aggregates and for adaptive, closed-loop process control to maintain key culture parameters. • Design and demonstrate a sterile, high-throughput sorting module with integrated AI-based characterization to homogenize batches by size, shape, and morphology at the start of culture or after completion of 3D cell model cultivation. • Implement automated medium exchange strategies driven by sensor data to reduce manual interventions and improve process consistency. • Establish data interoperability with Quality Systems (QS) and standardized data interfaces to enable traceability and regulatory readiness. • Validate the technology with iPSC-derived 3D cell models and demonstrate transferability to a Korean SME's application, enabling in-country customization, manufacturing, and market deployment.			

R&D Target Specifications	• Inline sensing (inside the bioreactor):
	  ○ Temperature control: ±0.1 °C; range 20–50 °C.
	  ○ pH sensing: ±0.1 pH units; range 6–8.
	  ○ Dissolved oxygen (DO): ±1% saturation; range 0–100%.
	  ○ CO_2: ±0.2% absolute; range 2–10% as applicable to culture conditions.
	• Online optical quality control (3D cell aggregates):
	  ○ Resolution sufficient to resolve 50–1000 µm diameter cell aggregates.
	• At-line AI-based classification and sorting:
	  ○ Sorting throughput: 6,000–12,000 aggregates/hour.
	  ○ Sorting purity: >90% of targeted parameter class.
	  ○ Post-sort viability of aggregates: >85%.			

R&D Plan	**Approach** • Maintain a risk-balanced, milestone-driven, design-build-test-learn approach with iterative hardware-software co-design. • Run parallel development tracks for sensors, imaging/AI, sorting, and automation, followed by system integration and biological validation. • End with a market- and regulatory-readiness demonstration and a structured knowledge transfer path to the Korean SME. **Time Horizon and Periods** (preliminary, will be discussed with the Korean SME) • Total duration: 36 months (3 years). • Period I (Months 0–12): Requirements capture, architecture finalization, and initial proofs-of-concept. Application-specific adaptation of already available inline sensor hardware, bioreactor interfaces, and initial online QC imaging plus AI prototypes; establish data interfaces and baseline QS alignment. • Period II (Months 13–24): System integration of core subsystems (WP1–WP4) and mid-term demonstrations. Validate AI-based classification and automated medium exchange, and advance data interoperability.			

- Period III (Months 25–36): Full end-to-end system demonstration with iPSC-derived 3D models; finalize transfer package for the Korean SME, finalize application-specific tailoring, and complete market-readiness and exploitation planning.

Milestones and Timeline (illustrative, will be discussed with the Korean SME)

- Milestone 1 (Month 12): End of Period I – Proof-of-concept for inline sensing, online QC imaging, and initial AI classification; architecture and interfaces locked for Period II.
- Milestone 2 (Month 24): End of Period II – Integrated subsystems demonstrated (WP1–WP4); automated medium exchange validated; data interoperability validated.
- Milestone 3 (Month 36): End of Period III – Full demonstration with iPSC-derived 3D models; technology transfer plan to the Korean SME finalized; market-readiness and exploitation plan established.

Work Packages (high-level, will be discussed with the Korean SME)

- WP0 – Project Management, Requirements & Architecture
- WP1 – Inline Sensor Hardware & Bioreactor Interface
- WP2 – Online QC Imaging & AI-Based Analysis
- WP3 – At-Line AI-Based Classification & Sorting
- WP4 – Automated Medium Exchange & Process Control
- WP5 – Data Interoperability & QS Interfaces
- WP6 – System Integration & Validation with iPSC 3D Models
- WP7 – Transfer to Korean SME & Demonstration for Market Readiness
- WP8 – Dissemination, IP & Exploitation Planning

Deliverables (will be discussed with the Korean SME)

- Modular sensor, imaging, sorting, and automation prototypes (WP1–WP4)
- AI models and real-time analytics pipeline (WP2, WP3)
- QS-ready data interfaces and interoperability package (WP5)
- Integrated demonstrator platform with iPSC-derived 3D models (WP6)
- Transfer package and SME-tailored configuration (WP7)
- (joint) IP, publications, exploitation plan (WP8)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Research scope • Develop a sensor-based bioreactor platform for quality-assured production and at-line AI-driven sorting of (complex iPSC-derived) 3D cell models. The system will be tailored for meeting customer specifications defined by the Korean SME. • The platform shall be suited for subsequent exploitation by the Korean SME by marketing it within Korea. The specific division of tasks between the Fraunhofer IBMT and the Korean SME can only be specified in a meaningful way once the Korean SME, the application addressed by the Korean SME and the technical capabilities of the Korean SME are known. In principle, Fraunhofer IBMT can provide the following resources and perform the following tasks: Resources to be provided • Access to cell culture laboratories, bioreactor systems, imaging hardware, and high-performance computing resources. • Bio-Engineering and technical-Engineering personnel (cell culture, hardware, software, AI/ML, data management) and project management support. • Experimental facilities for bench-scale testing and demonstration with iPSC-derived 3D models. Technical contributions and specific tasks • Define system architecture, verification and validation plans, and performance targets for submodules (sensors, AI-based classification, sorting). • Architecture and requirements finalization; component selection; and risk assessment. • Adapt and validate inline sensor hardware, online QC imaging, and at-line AI-driven classification/sorting algorithms, ensuring seamless hardware–software co-design. • Hardware–software integration, sensor calibration, and firmware development for inline and online sensing. • Implement automated medium exchange driven by sensor data to reduce manual interventions and improve process consistency. • Establish demonstrators and perform validation experiments with iPSC-derived 3D cell models to quantify improvements in batch homogeneity, and sorting accuracy. • Development of AI models for non-invasive evaluation and adaptive process control; creation of data pipelines and APIs for QS interfacing. • Knowledge transfer activities and preparation of technical materials for the Korean SME.
Expected R&R of Korean SME Partner	Research scope • Develop a sensor-based bioreactor platform for quality-assured production and at-line AI-driven sorting of (complex iPSC-derived) 3D cell models. The system will be tailored for meeting customer specifications defined by the Korean SME. • The platform shall be suited for subsequent exploitation by the Korean SME by marketing it within Korea. • The emphasis of the Korean SME will be on the adaptation and customization of the integrated platform for the targeted Korean application and preparations for its future commercialization as well as on validating regulatory compliance, safety, and reliability in the Korean context. The specific division of tasks between the Fraunhofer IBMT and the Korean SME can only be specified in a meaningful way once the Korean SME, the application addressed by the Korean SME

	and the technical capabilities of the Korean SME are known. The following information concerning resources to be provided and tasks to be performed by the Korean SME is provided for illustrative purposes only: Resources to be provided • Provide facilities and personnel for integration, customization, and testing activities. • Provide personnel which is familiar with the addressed application, particularly with regard to defining specifications from the user's perspective, regulatory requirements, manufacturing aspects for a future product, and market knowledge. Technical contributions and specific tasks • Define user specifications for the addressed application • Translate research outcomes into a market-ready device configuration tailored to a specific SME application. • Lead adaptation, customization, and prepare for future commercialisation of the integrated platform for the targeted Korean application. • Take care of regulatory alignment, and integration with local standards and customers. • Ensure interoperability with Quality Systems (QS) and standardized data formats. • Collaborate with the research institute to refine hardware/software interfaces for manufacturability and serviceability. • Participate in the development of AI models for non-invasive evaluation and adaptive process control as well as in creation of data pipelines and APIs for QS interfacing. • Generate and curate large-scale datasets for AI model development, training, and benchmarking. • Validate regulatory compliance, safety, and reliability aspects in the Korean context. • Share test data, user feedback, and market requirements to continuously steer development. • Act as a multiplier in Korea, promoting technology transfer, demonstrations, and early adoption among potential customers. • Develop the go-to-market plan, pricing, and distribution strategy for the Korean market. • Manage IP exploitation in Korea (in coordination with Fraunhofer IBMT).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• A validated, modular platform (or integrated system) for AI-assisted, sensor-based culture, monitoring, sorting, and data integration for 3D cell models, e.g. those derived from iPSCs. • Prototypes suitable for demonstration in an application relevant to Korea, with a clear pathway to commercialization by a Korean SME. • Transfer of knowledge and technology, enabling Korea-based production and market introduction, while leveraging Fraunhofer IBMT's scientific excellence and methodological standards. • Intellectual property potential around modular sensor systems, AI-based classification, and integrated sterile sorting workflows, with defined licensing or joint development arrangements.

6. References

Examples for (un-sensorized) Bioprocesses

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Balsters, J.M., Bäumchen, A., Roland, M., Diebels, S., Neubauer, J.C., Gepp, M.M., and Zimmermann, H. 2025. Drying of Functional Hydrogels: Development of a Workflow for Bioreactor-Integrated Freeze-Drying of Protein-Coated Alginate Microcarriers for iPS Cell-Based Screenings. Gels 11, 6, 439.

Fraunhofer Center for Sensor Intelligence ZSI

https://www.zsi.fraunhofer.de/en/Topics.html

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization